Sri Lanka Institute of Information Technology



Lab Submission
<Lab sheet 02>

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Discrete Mathematics | IT1160

B.Sc. (Hons) in Information Technology

Part 01)

01)

```
In [5]: cities =["Galle" , "Colombo" ,"Matara"]
        city = input("Enter a city: ")
        cities.append(city)
        print("Updated cities" , cities)
```

```
Enter a city: kandy
Updated cities ['Galle', 'Colombo', 'Matara', 'kandy']
```

02)

```
In [10]: A = (1,2,3,4,5)
         print(A[0])
         print(A[4])
         print(A[2])
```

```
1
5
3
```

```
In [11]: A =(1,2,3,4,5)
         A = list(A)
         A[4] = 10
         A = tuple(A)
         print(A)
```

```
(1, 2, 3, 4, 10)
```

05)

```
In [52]: 25 -15 = 10
         22 - 15 = 7
         25+22-15 = 32
         40-32 = 8

           Cell In[52], line 1
             25 -15 = 10
             ^
         SyntaxError: cannot assign to expression here. Maybe you meant '==' instead
         of '='?
```

03)

```
In [18]: book = {"title": "1984", "author": "George Orwell"}
         book["Year"]  = 1949
         print(book)

         book["author"] = "George Arthur Orwell"
         print(book)

         print(book["Year"])

         book.pop("Year")
         print(book)

         {'title': '1984', 'author': 'George Orwell', 'Year': 1949}
         {'title': '1984', 'author': 'George Arthur Orwell', 'Year': 1949}
         1949
         {'title': '1984', 'author': 'George Arthur Orwell'}
```

04)

In [25]:
```
A = {"U1", "U2", "U3", "U4", "U5", "U6"}
B = {"U3", "U4", "U5", "U7"}
C = {"U5", "U7"}

print(C.issubset(A))

A.intersection(B)
```

```
False
```

Out[25]:
```
{'U3', 'U4', 'U5'}
```

In [26]:
```
B.difference(C)
```

Out[26]:
```
{'U3', 'U4'}
```

In [27]:
```
print(A.isdisjoint(C))
```

```
False
```

In [28]:
```
A.union(B)
```

Out[28]:
```
{'U1', 'U2', 'U3', 'U4', 'U5', 'U6', 'U7'}
```

In [30]:
```
A.symmetric_difference(B)
```

Out[30]:
```
{'U1', 'U2', 'U6', 'U7'}
```

Part 02)

01)

In [62]:
```
scores = [78, 85, 95, 62, 88, 75, 95, 85, 62, 90]

unique_scores = []

for score in scores:
    if score not in unique_scores:
        unique_scores.append(score)

print(unique_scores)
unique_scores.pop(0)
print(unique_scores)
unique_scores.append(99)
print(unique_scores)

top3 = sorted(unique_scores, reverse=True)[:3]

print("Top 3 scores:", top3)
```

```
[78, 85, 95, 62, 88, 75, 90]
[85, 95, 62, 88, 75, 90]
[85, 95, 62, 88, 75, 90, 99]
Top 3 scores: [99, 95, 90]
```

02)

In [66]:
```
for product in products.values():
    if product[1] == "Furniture" and product[3] > 3000:
        print(product[0])
```

```
Table
```

In [67]:
```
for product_id, product in products.items():
    if product[2] < 50:
        print(product_id)
```

```
101
102
104
107
```

In [65]:
```
products = {
    101: ["Laptop", "Electronics", 15, 55000],
    102: ["Table", "Furniture", 30, 4500],
    103: ["Pen", "Stationery", 100, 20],
    104: ["Mobile Phone", "Electronics", 25, 15000],
    105: ["Chair", "Furniture", 50, 2500],
    106: ["Notebook", "Stationery", 200, 50],
    107: ["Headphones", "Electronics", 40, 2500],
    108: ["Water Bottle", "Kitchenware", 60, 350]
}

for product in products.values():
    if product[1] == "Electronics":
        print(product[0])
```

```
Laptop
Mobile Phone
Headphones
```

03)

In [73]:
```
sale_event_a = {1001, 1002, 1003, 1004, 1005, 1006}
sale_event_b = {1004, 1005, 1006, 1007, 1008, 1009}

print(sale_event_a & sale_event_b)

print(sale_event_a - sale_event_b)

print(sale_event_b - sale_event_a)

print(sale_event_a | sale_event_b)

print(sale_event_a ^ sale_event_b)
```

```
{1004, 1005, 1006}
{1001, 1002, 1003}
{1008, 1009, 1007}
{1001, 1002, 1003, 1004, 1005, 1006, 1007, 1008, 1009}
{1001, 1002, 1003, 1007, 1008, 1009}
```

04)

In [77]:
```
fever = {201, 202, 203, 204, 205, 206, 207}
cough = {203, 204, 205, 208, 209, 210}
fatigue = {202, 204, 206, 209, 211, 212}

print(fever & cough & fatigue)


print(fever - cough - fatigue)

two_or_more = (fever & cough) | (fever & fatigue) | (cough & fatigu

print(two_or_more)
print(len(two_or_more))

priority = fever & (cough | fatigue)

print(priority)
```

```
{204}
{201, 207}
{209, 202, 203, 204, 205, 206}
6
{202, 203, 204, 205, 206}
```

05)

In [80]:
```
F = {"React", "Angular"}
B = {"Node.js", "Django"}

import itertools

F = {"React", "Angular"}
B = {"Node.js", "Django"}

pairs = list(itertools.product(F, B))

print(pairs)

D = {"Docker", "Nginx"}


import itertools

D = {"Docker", "Nginx"}

subsets = []

for r in range(len(D) + 1):
    subsets.extend(itertools.combinations(D, r))

print(subsets)
```

```
D = {"Docker", "Nginx"}

subsets = []

for r in range(len(D) + 1):
    subsets.extend(itertools.combinations(D, r))

print(subsets)
```

```
[('Angular', 'Node.js'), ('Angular', 'Django'), ('React', 'Node.js'), ('Rea
ct', 'Django')]
[(), ('Docker',), ('Nginx',), ('Docker', 'Nginx')]
```